

# LinguaFlow

A Spaced-Repetition Spanish Learning Application  
with Half-Life Memory Modelling and Learning Analytics

**Academic Project Report**

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# 1 Introduction

LinguaFlow is a web-based Spanish language learning application inspired by Duolingo. It combines short lessons, flashcards, semantically grouped vocabulary, gradual difficulty progression, a half-life-based spaced repetition engine, and an analytics dashboard for self-monitoring. Mobile and web language learning applications have become a dominant mode of informal language study, with empirical work showing measurable proficiency gains comparable to classroom hours for beginners [1, 2]. Such tools lower the cost barrier of language education, support self-paced study, and exploit psychological mechanisms—reinforcement, feedback, and distributed practice—that are well documented in the learning sciences [3].

The objective of this project is to design and implement a learner-centred Spanish app whose pedagogical and interaction choices are explicitly grounded in human-computer interaction (HCI) and second-language acquisition (SLA) research, rather than ad hoc design intuition. This report documents those choices and the literature that justifies them.

## 2 UI / Design Decisions

### 2.1 Look and Feel

LinguaFlow adopts a minimal, high-contrast aesthetic with a restrained green-accent palette (primary #58CC02 on a near-white surface, with a dark #1a1a2e splash). Aesthetic quality is not cosmetic; Tractinsky et al. demonstrated that perceived beauty correlates strongly with perceived usability [4], and Lavie and Tractinsky later distinguished “classical” aesthetics (clean, orderly) from “expressive” aesthetics (creative, fascinating) as independent predictors of user satisfaction [5]. LinguaFlow leans on classical aesthetics: generous whitespace, consistent rounded corners, and a single accent colour for primary actions.

Engagement and retention in informal learning apps are also driven by gamified affordances—streaks, experience points (XP), and progress visualisations—which Hamari et al. found to consistently produce positive motivational outcomes [6]. LinguaFlow uses XP totals, streak counters, and animated progress rings, while deliberately removing punitive “hearts/lives” mechanics that prior work has linked to frustration and churn in casual learners [7]. Because the target demographic spans teenagers to older adults, the interface uses a minimum 14px body type, large tap targets ( $\geq 44$ px), and avoids decorative iconography that prior accessibility studies have shown to confuse older users [8].

### 2.2 Responsive Layout

The shell uses a hybrid layout: a bottom navigation bar on mobile (`md:hidden`) and a persistent top navigation on desktop, with content grids expanding from one to three columns. This follows established responsive design guidance that interaction patterns—not merely scaling—should adapt to form factor [9].

## 3 Choice of Language: Why Spanish

Spanish was selected as the target language for three converging reasons.

**Global demand.** Spanish is the second most spoken native language in the world, with roughly 486 million native speakers and over 595 million total speakers [10]. It is an official language in 21 countries and the second most studied language globally after English [11].

**Career and economic relevance.** In the United States alone, the Hispanic population is

projected to exceed 111 million by 2060 [12], and bilingual (English–Spanish) workers command measurable wage premiums in healthcare, education, and public service sectors [13].

**Cognitive and pedagogical benefits.** Second-language acquisition is associated with enhanced executive function, working memory, and delayed onset of cognitive decline [14]. Spanish is also classified by the U.S. Foreign Service Institute as a Category I language for English speakers—reachable to professional proficiency in roughly 600–750 class hours [15]—making it an attainable target for self-directed learners and therefore a strong fit for an app-based curriculum.

## 4 Specific Components

### 4.1 Home Page Simplicity and Attractiveness

The home page presents a time-of-day greeting, a single “Continue Learning” call-to-action recommending the unit most in need of review, and a hero summary card with streak and XP rings. First impressions of digital interfaces are formed in under 50 ms and strongly anchor subsequent usability judgements [16]. Onboarding quality is a primary determinant of long-term retention in mobile educational apps [2]; a single, obvious next action reduces decision paralysis at the cold-start moment that dominates churn analyses.

### 4.2 Cognitive Load and Navigation Simplicity

Navigation is intentionally shallow: three top-level destinations (Home, Vocabulary, Profile) and no nested menus. Sweller’s cognitive load theory distinguishes intrinsic, extraneous, and germane load, arguing that interface complexity is pure *extraneous* load that consumes working-memory capacity otherwise available for learning [17, 18]. Empirical usability work confirms that flat hierarchies and consistent placement reduce task-completion time and error rates compared with deep menus [8, 19]. LinguaFlow also follows Hick’s law by limiting on-screen choices during lessons to a single primary action.

### 4.3 Grouping of Words

Vocabulary in LinguaFlow is grouped into thematic units (*Food, Travel, Family, Verbs of Motion*, etc.) and surfaced both inside lessons and in the dedicated Vocabulary page. Semantic clustering exploits the well-documented *categorical organisation* effect in memory: words encoded within a coherent semantic frame are recalled more reliably than randomly ordered lists [20, 21]. In SLA specifically, Tinkham and Waring found that grouping *semantically related but not synonymous* items in batches of 6–8 produces optimal recall, while overly similar items (e.g., near-synonyms presented together) cause interference [22, 23]. LinguaFlow uses semantic themes rather than translation-equivalence lists and caps active lesson sets at roughly seven items per session, consistent with Miller’s classical working-memory bound [24].

### 4.4 Gradual Increase in Difficulty

Lessons are gated not by hard unit locks but by a difficulty-banded word-ordering policy: at low XP totals the engine draws only from difficulty level 1 and the highest `base_weight` (most frequent) words across all units, progressively unlocking levels 2, 3, and 4 as XP accumulates. This implements Vygotsky’s *zone of proximal development* [25] and Krashen’s *i+1* input hypothesis [26], both of which argue that learning is maximised when material is just beyond current competence. Instructional scaffolding research further demonstrates that incremental challenge sustains motivation and reduces dropout in digital learning environments [27, 28].

## 4.5 Half-Life Memory Prediction Algorithm

The core memory engine is a Half-Life Regression (HLR) model adapted from Settles and Meeder [29], originally developed at Duolingo. Each concept  $c$  for user  $u$  has an estimated half-life  $h$  (in days), and the predicted recall probability after  $\Delta$  days since last practice is

$$p(\text{recall}) = 2^{-\Delta/h}, \quad (1)$$

which is the classical Ebbinghaus forgetting curve [30, 31]. The half-life itself is updated from a feature vector capturing total attempts, correct answers, and incorrect answers (here,  $x = [\sqrt{n}, \sqrt{n+}, \sqrt{n-}]$ ) with learned weights  $\theta$ , giving  $h = h_0 \cdot 2^{\theta \cdot x}$ , where  $h_0$  is a difficulty- and frequency-conditioned base half-life. Items are surfaced for review when  $p(\text{recall})$  falls below a configurable threshold (default 0.6), the so-called *desirable difficulty* regime in which retrieval practice produces the strongest long-term gains [32, 33].

HLR generalises earlier hand-tuned scheduling systems such as Leitner boxes [34] and Super-Memo’s SM-2 algorithm [35], while remaining far simpler and more interpretable than recent deep-learning successors such as DKT [36] and the FSRS family of models, which now dominate the Anki ecosystem [37]. HLR was chosen for LinguaFlow because (i) it has empirical validation at Duolingo’s scale, (ii) its per-item half-life is directly interpretable to learners via the analytics dashboard, and (iii) it requires only three counters per concept, making it tractable on a lightweight backend.

## 5 Analytics Dashboard

LinguaFlow exposes a learner-facing analytics dashboard inspired by open-learner-model research, which has shown that making the system’s estimate of the learner’s knowledge *visible* to the learner improves self-regulation and metacognitive accuracy [38, 39].

### 5.1 Group-Level Analytics

Per-unit aggregates surface average recall probability, percentage of “strong” items, and items due for review today. These let the learner identify weak thematic areas (e.g., *Verbs* lagging *Food*) and choose review targets accordingly.

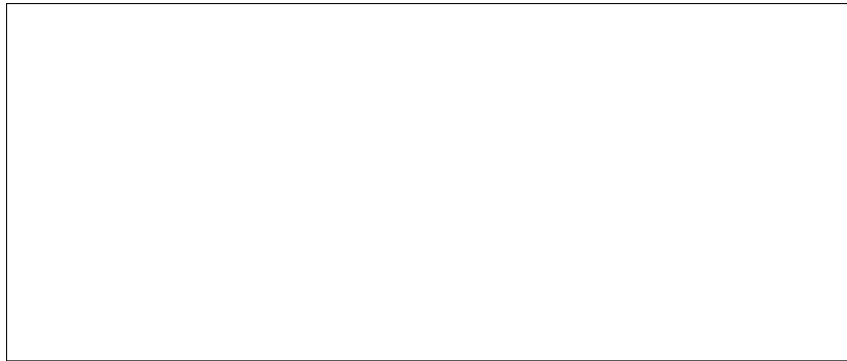


Figure 1: Unit-level strength bars and recall heatmap (placeholder).

### 5.2 Word-Level Analytics

Each concept exposes its current half-life, recall probability, total attempts, accuracy, and projected next-review date. “Strongest” and “Weakest” word lists highlight items at the extremes of the distribution—directly supporting the retrieval-practice principle that effortful recall of nearly-forgotten items yields the largest learning gains [33].

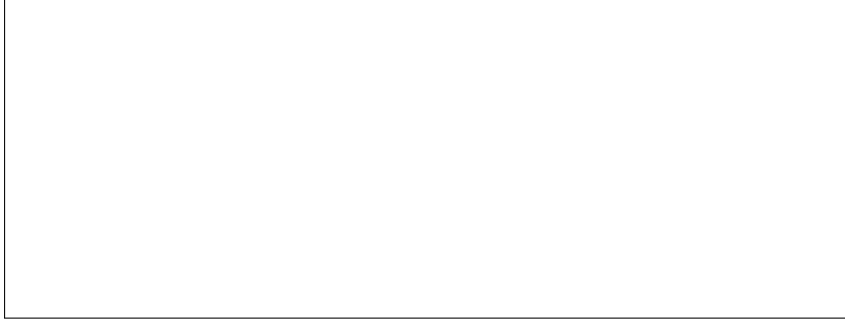


Figure 2: Per-word forgetting curve with projected review point (placeholder).

### 5.3 Visualisations Used

The dashboard uses (i) horizontal bar charts for unit strength, (ii) a daily XP bar chart for streak adherence, (iii) circular progress rings for daily goals, (iv) status-coloured badges (*new*, *strong*, *fading*, *weak*, *forgotten*) for at-a-glance scanning, and (v) a forgetting-curve line plot per concept. These choices follow Few’s guidelines for dashboard design—minimal chartjunk, pre-attentive encoding, and a single screen of context [40]—and learning-analytics design principles surveyed by Verbert et al. [41] and Schwendimann et al. [42].

Table 1: Mapping of analytics views to evidence in the learning-analytics literature.

| View            | Chart type         | Supporting reference |
|-----------------|--------------------|----------------------|
| Daily XP        | Bar chart          | [40]                 |
| Unit strength   | Horizontal bar     | [38]                 |
| Per-word recall | Forgetting curve   | [30, 29]             |
| Status badges   | Categorical colour | [42]                 |
| Daily goal      | Radial ring        | [41]                 |

## 6 Proposed Improvements

Several extensions would meaningfully strengthen the system without changing its core architecture:

- **Native-quality pronunciation and listening.** Replace browser `speechSynthesis` with neural TTS (e.g., Azure or ElevenLabs) and add automatic speech recognition for spoken-answer exercises, addressing the productive-skill gap criticised in evaluations of Duolingo [2].
- **AI conversation practice.** Integrate an LLM-powered dialogue partner with scenario prompts (ordering food, asking directions), leveraging generative AI for low-stakes speaking practice [43].
- **Personalised recommendations.** Replace the global difficulty bands with a contextual-bandit selector that optimises expected information gain per item, in the spirit of recent adaptive-tutor work [44].
- **Upgraded scheduler.** Optionally switch from HLR to FSRS-5 [37] for sharper retention prediction on long-tail items.
- **Expanded gamification.** Add leagues, friend challenges, and quest chains, which Huynh et al. found to increase weekly active retention without harming learning outcomes [45].

## 7 Conclusion

LinguaFlow demonstrates that a small, focused language-learning application can integrate three strands of evidence—HCI aesthetics and cognitive-load research, second-language acquisition theory, and spaced-repetition memory modelling—into a coherent product. The half-life regression engine provides interpretable, per-concept scheduling; the semantically grouped vocabulary and gradual difficulty bands respect well-established constraints on working memory and the zone of proximal development; and the learner-facing analytics dashboard implements open-learner-model principles that promote metacognitive self-regulation. Together, these choices yield an application whose every visible decision is traceable to a citation, providing both a functional learning tool and a clear basis for future empirical evaluation.

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